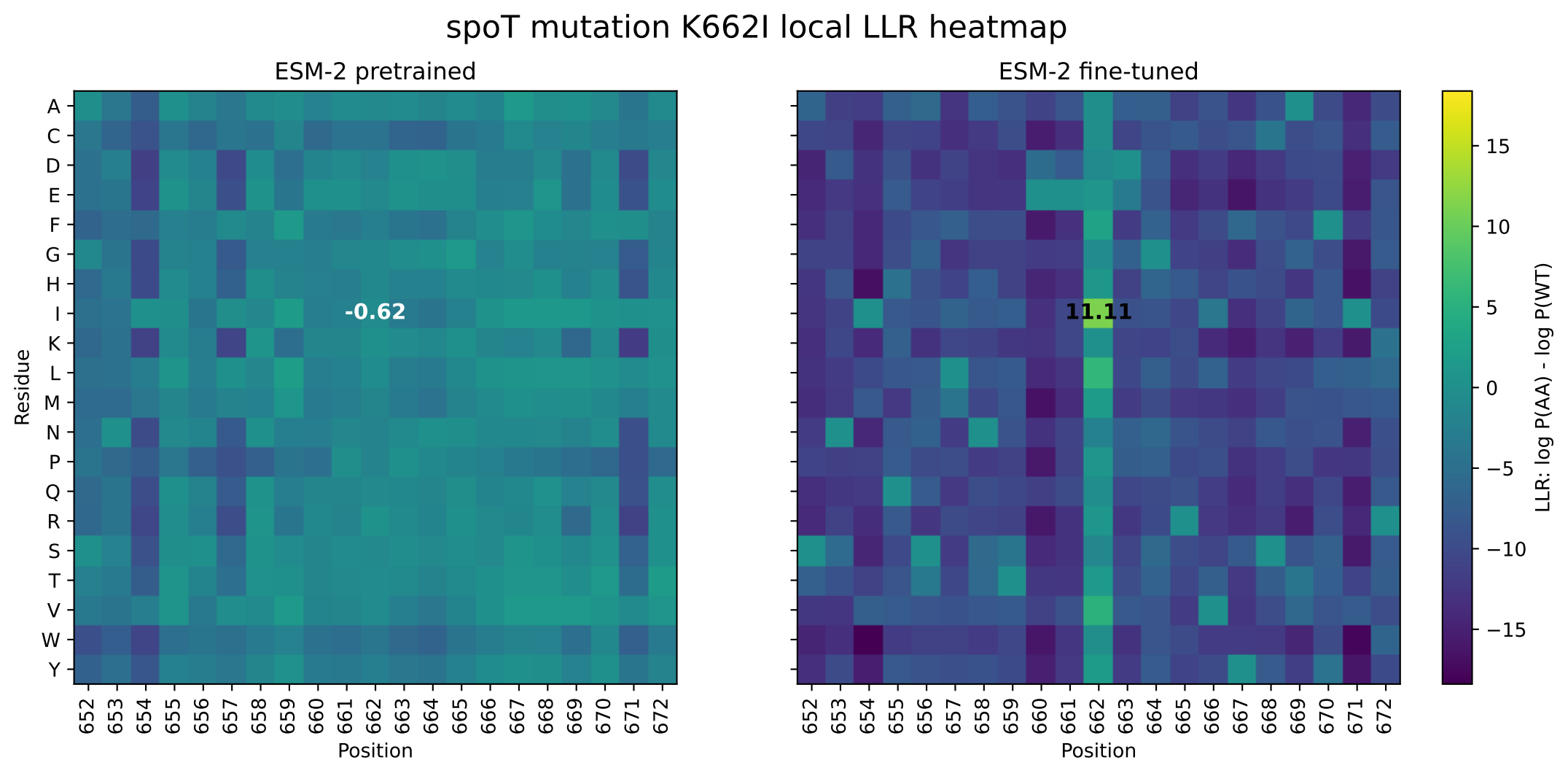
spoT mutation K662I local LLR heatmap
ESM-2 pretrained
ESM-2 fine-tuned
-0.62
11.11
Residue
A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
Position
15
10
5
0
−5
−10
−15
LLR: log P(AA) - log P(WT)